INDIA INNOVATES 2026

TEAM NAME :
GRID HUNTERS

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PROBLEM STATEMENT

Cyber Threats Are Outpacing Defenses

- India ranked 10th globally in cybercrime with ₹1.7 lakh crore lost in 2023
- IoT cyberattacks grew 400% in 2023 most targeting unprotected sensor networks
- 95% of cyberattacks go undetected until data is already stolen

Major Grid Attacks :

- **Grid Attacks during Operation Sindoor (India)**
- **Mumbai (2020) Sabotage attack.**
- **Grid Attack during Russia-Ukraine War (2015/2016/2018/2019/2022)**

CORE PROBLEMS:

DATA TRAVELING FROM SENSOR TO SERVER CAN BE INTERCEPTED AND MODIFIED

CURRENT SYSTEMS HAVE NO WAY TO DETECT IF SENSOR DATA WAS TAMPERED

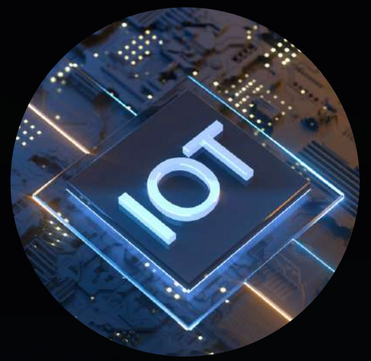
ALL CURRENT SYSTEMS RELY ON ONE INTERNET CONNECTION

ATTACK ON ISP PATHWAY LEADS TO COMPLETE SYSTEM SHUTDOWN

TRADITIONAL FIREWALLS ARE REACTIVE THEY BLOCK ONLY AFTER BREACH HAPPENS

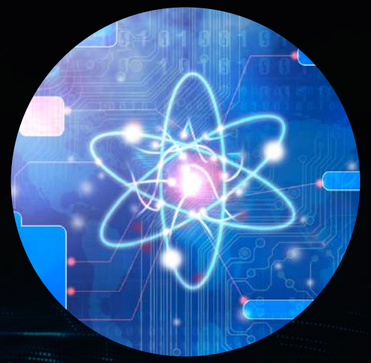
BY THE TIME FIREWALL BLOCKS DATA IS ALREADY ACCESSED

IF A DEVICE IS COMPROMISED, IT CAN LEAD TO GRID INSTABILITY, INCORRECT GRID DECISIONS OR, EVEN A LARGE SABOTAGE AND CITY BLACKOUTS



SOLUTION:

SMART SHIELD



SMART SHIELD HAS THREE PILLARS:

DECEIVE: Honeypot Windows OS lures attackers into a fake system so they never reach real data

VERIFY: Every data packet passing through IoT nodes is hash-verified before reaching main system tampered data is caught instantly

ISOLATE: Breaking Point Logic automatically severs all attack pathways and switches to backup ISP zero human intervention, zero downtime

PROJECT OVERVIEW

SYSTEM ARCHITECTURE - 3 PROTECTION ZONES

HONEYPOT : THE DECEPTION CORE

WHAT IT IS:

A DECOY COMPUTER THAT LOOKS IDENTICAL TO THE REAL SYSTEM PLACED IN THE ATTACKER'S PATH DELIBERATELY

ESP1-A COMPLETELY INDEPENDENT ESP THAT READS AND SENDS BOTH HASH AND ROW DATA TO VERIFICATION SYSTEM

ESP2- AN ESP32 DEDICATED TO CONVERTING RAW SENSOR VOLTAGE INTO A CRYPTOGRAPHIC HASH CODE

ESP 3 - AN ESP32 DEDICATED TO SENDING THE ORIGINAL UNMODIFIED VOLTAGE READING TO THE VERIFICATION SYSTEM

ESP8266 ± ZMPT101B — PHYSICAL LAYER SECURITY

WHAT IT IS:

A PHYSICAL VOLTAGE SENSOR THAT MONITORS THE POWER SUPPLY OF THE HARDWARE INFRASTRUCTURE ITSELF

HOW IT WORKS:

ZMPT101B SENSOR CONNECTS DIRECTLY TO VOLTAGE SOURCE

READS AC MAINS VOLTAGE CONTINUOUSLY

ESP8266 CONVERTS ANALOG READING TO DIGITAL AND SENDS TO MAIN SYSTEM VIA MQTT

IF VOLTAGE DROPS BELOW 200V THRESHOLD → PHYSICAL TAMPERING ALERT FIRES

MAIN SYSTEM DISPLAYS LIVE VOLTAGE GRAPH IN REAL TIME

WHAT THE SYSTEM IS:

SMARTSHIELD IS A CYBER-PHYSICAL SECURITY ARCHITECTURE THAT PROTECTS A MAIN COMPUTER SYSTEM FROM CYBERATTACKS

SmartShield works in five simultaneous layers. ISP_1 feeds all incoming traffic into a honeypot decoy so attackers are trapped before they get anywhere near real data.
Three independent ESP32 nodes read the same sensor and cross-verify each other's data using cryptographic hashing and if any tampering in transit is caught within one second.
A physical voltage sensor monitors hardware integrity independently of all software. A Windows verification system acts as central judge for only forwarding data that passes all checks,And the main system sits completely isolated and protected by firewall whitelisting, port knocking, and VPN encryption. The moment any layer detects a threat the Breaking Point Logic fires, all attack pathways are severed, and the system switches to backup ISP_3 with zero seconds of downtime.

WHAT WE BUILT:

FULLY WORKING PHYSICAL PROTOTYPE

REAL ESP32 NODES SENDING LIVE MQTT DATA

HASH VERIFICATION RUNNING IN REAL TIME

BREAKING POINT LOGIC TESTED AND DEMONSTRATED

LIVE VOLTAGE MONITORING VIA ZMPT101B

KEY ADVANTAGES:

Proactive not reactive : It traps attacker before they reach real system

Multi-layer : It has 5 independent protection mechanisms simultaneously

Self-healing : system recovers automatically without human action

Attacker intelligence : honeypot logs reveal attacker's tools and methods

Affordable : deployable at 10x lower cost than enterprise solutions

THANK YOU!